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ODD END SEMESTER EXAMINATION DECEMBER - 2025

Name of the Course: B. TECH (ME)

Semester: V SEM.

Name of the Paper: SUPPLY CHAIN MANAGEMENT

Paper Code: **TME 515**

Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	Define Supply Chain Management (SCM). Discuss its main goals and relevance in today's competitive business environment.	CO1,CO2
(b)	Identify and explain the key drivers of supply chain performance. How do these drivers influence operational efficiency and responsiveness?	
(c)	Describe the various types of flows within a supply chain. Give suitable examples to illustrate how these flows maintain coordination among partners.	
Q2	(20 marks)	
(a)	Define facility location. Discuss the important factors that affect location decisions and capacity allocation while designing a supply chain network.	CO2,CO3
(b)	What is forecasting? Explain the different forecasting approaches and describe how forecasting supports effective supply chain planning.	
(c)	Explain the concept of Sales and Operations Planning (S&OP). Outline its steps and discuss how it helps in balancing demand and supply.	
Q3	(20 marks)	
(a)	What is inventory? Discuss the major types of inventory and explain the Economic Order Quantity (EOQ) model in detail with assumptions and formula.	CO2,CO4
(b)	Differentiate between centralized and decentralized inventory systems. Discuss how each approach affects service level and operational cost.	
(c)	Describe the main modes of transportation used in logistics. How do transportation choices and costs influence supply chain performance?	
Q4	(20 marks)	
(a)	Define strategic sourcing. Discuss the criteria and methods for supplier selection and evaluation in a competitive global environment.	CO3,CO5
(b)	What are strategic partnerships in supply chain management? Explain their importance in improving collaboration and achieving long-term efficiency.	
(c)	What is e-procurement? Discuss its benefits, challenges, and role in shaping sourcing strategies and make-or-buy decisions.	
Q5	(20 marks)	
(a)	Explain the role of Information Technology (IT) in modern supply chains. How do tools like ERP, RFID, and Blockchain improve visibility and integration?	CO4,CO6
(b)	What is sustainable supply chain management? Discuss the role of green and reverse logistics in achieving environmental and social goals.	
(c)	Define supply chain integration and collaboration. Support your explanation with real-world examples or case studies of successful supply chain practices.	

